Akash Kothari

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Computer Engineering (Co-op) — *GPA: 4.0/4.0*

Experience

Software Engineer Intern - Orders & Returns — Sep. 2026 - Present
Shopify — *Toronto, ON*

- Closed **4** authorization gaps in the Returns **GraphQL** API that let apps access other apps' return data, shipping each fix behind feature flags with production traffic validated in observability tooling.
- Implemented **ReBAC** (relationship-based access control) support for Returns, restoring access for **Amazon** fulfillment apps that were incorrectly blocked from their returns.
- Eliminated sequential GraphQL query waterfalls on high-traffic merchant-facing Returns pages by moving default-selection logic to the backend, cutting initial load time by ∼**50%** (∼**500ms**).

Software Engineer Intern — May 2026 - Sep. 2026
QuiikMart - Grocery Delivery Startup — *Remote*

- Developed a recommendation engine with a **nightly sweeper** that rolls purchase totals into a baseline popularity store, combined with weighted per-user signals and **Redis** caching for guest sessions.
- Added user event logging across **6** interaction types (e.g. views, add-to-carts, purchases) in a **NestJS/TypeScript** and **PostgreSQL** backend, capturing **500+** events/day as training data to learn the engine's signal weights.
- Created a bulk CSV product-upload system processing up to **1,000** products per upload, with server-side validation, flexible column aliasing, category matching, and a two-step validate-then-import flow.

ML Software Engineering Intern — Jan. 2026 - Apr. 2026
RamSoft — *Toronto, ON*

- Built a real-time clinical transcription pipeline fusing **MedASR** (dictation) and **WhisperFlow** (voice commands) via timestamp alignment and fuzzy matching, replacing an outsourced service with private in-house inference.
- Matched the outsourced vendor's accuracy benchmark, cutting word error rate by **80%** across **200** audio samples via chunked streaming and PyTorch confidence scoring; deployed on **Kubernetes** and piloted with radiologists.
- Developed a deterministic replay eval harness scoring normalized WER against reference transcripts, with regression checks to prevent overfitting, plus a multi-pass **LLM-as-judge** that flags and fixes transcript errors.
- Executed a **Cosmos DB** → **SQL Server** migration for scheduling data using **dual-writes** and backfill.
- Built an internal **MCP server** for Azure SQL with MFA token auth, enabling natural-language database queries.

Cloud Software Engineering Intern — Jun. 2025 - Sep. 2025
Tripleview Technologies — *Mississauga, ON*

- Automated Teams call-recording retrieval via **Microsoft Graph** and C# to feed an AI summarization pipeline.
- Converted **8+** transcripts/hour into structured **Zendesk** tickets via webhooks, removing **10+ hours/week** of manual work; shipped to production and adopted across internal teams.

Projects

Mercury Mortgages - Expense Automation | *React, FastAPI, Supabase, GPT-4o Vision, Redis*

- Built and solely maintain a multi-tenant expense platform used daily across **21** companies, processing **$674K+** in expenses and auto-matching **87.8%** of receipts to transactions, replacing a manual bookkeeping role.
- Engineered an email ingestion pipeline processing **600+** emails and **1,400+** receipts with **GPT-4o Vision** field extraction, using DB-tracked workers so no attachment is dropped on failure.
- Designed a rules engine mapping **573** vendors to **39** GL codes across **6** currencies, plus lazy PDF assembly with upload compression for reports exceeding **700** pages.

Self-Driving Car Simulator | *Unity, C#, PyTorch, Reinforcement Learning*

- Wrote a vehicle physics engine from scratch in Unity (no built-in wheel colliders): raycast spring-damper suspension, slip-based tire grip via tunable friction curves, and a configurable FWD/RWD/AWD drivetrain.
- Structured the sim as a reinforcement learning environment (raycast sensors, waypoint-based rewards) and implemented a **custom PPO training loop from scratch** (no ML-Agents) over a Python-Unity socket bridge.

Technical Skills

Languages: Python, C/C++, C#, TypeScript, JavaScript, Ruby, Java, SQL
Frameworks & Libraries: FastAPI, NestJS, Ruby on Rails, Node.js/Express, GraphQL, React, Next.js, Unity
Cloud & DevOps: Azure, Docker, Kubernetes, PostgreSQL, Redis, Supabase, CI/CD, Git, REST APIs
ML & Data: PyTorch, NumPy, Pandas, Reinforcement Learning, MCP, LLMs, LLM Evaluation